

Golden Rules for Designing Wearables



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Technology-aided behaviour analysis and modification has been there since four years now. However, earlier devices had the tendency to be more about delivering technology in a wearable device than about affecting behavioural change in the user.

Two researchers from Mayo Clinic Centre for Innovation had reviewed a smartband in 2011 on Mayo Blog. While they believed the technology would work, manual data syncing, poor battery life, lack of data privacy and wired connections all pointed towards a lacklustre hardware design.

Proof that wearable electronics have matured was on show at CES 2015. Both niche companies like Jawbone and tech giants like Samsung have learnt that awkward and clumsy designs do not appeal to anyone except geeks and early technology adopters.

While year 2014 saw a lot of start-ups in this area with most generating healthy business, 2015 will probably be the year when consumers decide the winners and losers.

Balancing act with the battery

Since wearables are usually small, the batteries included in these are also small and have limited capacity. Fortunately, there are ways to increase the life of a battery.

Marten Smith, marketing manager, medical products group, Microchip Technology, explains that microcontrollers (MCUs) with low run currents, multiple low-power sleep modes, flexible wake-up sources and intelligent power management are critical to the success of a wearable device. Reduced power consumption enables

the devices to use smaller batteries, run longer between recharging cycles and offer smaller product footprints.

Offloading data processing onto a cloud server or smartphone is an effective way to ensure that power-consuming calculations are done elsewhere, and the device itself focuses on what it does best, that is, data collection and, in some cases, presenting the results in an easy-to-understand way. When you offload the processing, it also means that you need not have a powerful processor on your device any more.

For instance, in the case of Microsoft Band and Samsung Gear Fit, online reviews reveal that Microsoft Band has a typical battery life of two days, while Samsung Gear Fit gets an average of three days. But Gear Fit has a battery that is almost double in size as compared to Microsoft Band.

Why this discrepancy? Although both run on ARM Cortex-M4 MCUs, Microsoft Band runs on Freescale's Kinetis K24 MCU, which has its central processing unit (CPU) clocked at 120MHz. On the other hand, Samsung's device runs an MCU at 180MHz. Factors like 3.6cm versus 4.6cm (1.4-inch versus 1.8-inch) screen size, display technology and sensors also affect battery life.

There is also the option of doing what Garmin did with their Vivofit, that is, powering it using a classic Panasonic CR1632 3V lithium-coin battery. This might, however, work only for lightly-featured devices since Vivofit runs on a Cortex-M0 based processor (Samsung and Microsoft devices run on Cortex-M4 processors).

What is the difference? A Cortex-M4 has single-instruction multiple-data (SIMD) and floating-point capabilities that let it process magnitudes faster than Cortex-M0, with the fallout being power consumption, which is magnitudes higher.

It is also possible to optimise the code running on an existing device to improve its performance or battery life. We first saw this being done by Elon Musk for his Tesla cars, when over-the-air (OTA) software

Breakup of Jawbone UP3

